

Assignment 1

This assignment will guide you through a Business Intelligence (BI) workflow using **AdventureWorksDW2022**. You will use Alteryx for ETL and data transformation, SQL Server (SSMS) for data storage and management, and Power BI for visualization and insights generation.

You will work with the following tables:

- **FactResellerSales** – Reseller sales transactions
- **FactInternetSales** – Online direct sales data
- **DimCustomer** – Customer demographics
- **DimProduct** – Product information
- **DimDate** – Calendar date references
- **DimGeography** – Regional information

Tasks to Complete:

1. Extract relevant tables from SQL Server (SSMS) using Alteryx.
2. Create the following measures in Alteryx:
 - **Total Sales Amount** → $\text{SUM}(\text{SalesAmount})$
 - **Total Orders** → $\text{COUNT}(\text{SalesOrderNumber})$
 - **Profit Margin** → $(\text{SUM}(\text{SalesAmount}) - \text{SUM}(\text{TotalProductCost}))$
 - **Average Order Value** → $(\text{Total Sales Amount} / \text{Total Orders})$
3. **Join** FactSales with DimCustomer, DimProduct.
4. **Store** the transformed dataset as a new table in SQL Server (FactSalesMeasures).
5. **Connect Power BI** to SQL Server and import the new FactSalesMeasures table.
6. Create at least four interactive visualization:
 - **Sales Trends Over Time** → Line chart with Total Sales Amount by Year.
 - **Top-Selling Products** → Bar chart showing Total Sales by Product Category.

The final report should include:

1. **Overview of Business Insights:** A summary of key findings from Power BI visualizations, including trends in sales, customer segmentation, and geographical performance.
2. **ETL Process Explanation:** A brief description of the transformations performed in Alteryx, including the measures created and how the data was structured for analysis in Power BI.
3. **Screenshots:** Include relevant screenshots of the **Alteryx workflow**, **SQL Server table**, and **Power BI dashboard** to support your explanations.